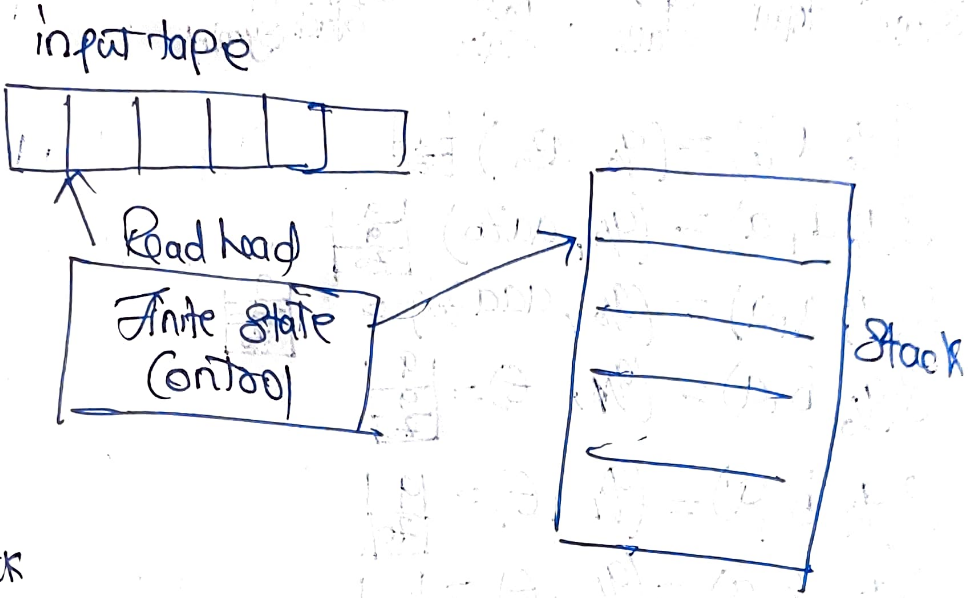
Module 04

Model of PDA (Push down Automata)

It has 3 components



check PDA not ← PDA Can be defined with 7 tuples:-

$M = (Q\ \Sigma\ \Gamma\ \delta\ q_0\ Z_0\ F)$

- Q - Set of finite State
- Σ - Input alphabet
- Γ - Stack alphabet
- q_0 - Starting State
- Z_0 - top of the Stack
- F - Final State
- δ - Transition State

$$\delta \rightarrow \underbrace{Q \times (\Sigma \cup \{\epsilon\})}_{\text{I/o tape}} \times \underbrace{\Gamma}_{\text{Stack}} \rightarrow Q \times \Gamma^*$$

Γ^* - a String of the Symbol that replaces the Stack.

Q- Construct PDA for the language $L = \{a^n b^n / n \geq \ldots\}$

$\Rightarrow L = \{ab, aa, bb, aaaabbbb, aabb - -\}$

$\delta(q, a, X) = (p, V)$

- q: Current State
- a: Input
- X: top of Stack
- p: New State
- V: State of Stack Symbol

$\delta(q_0, a, Z_0) = (q_0, aZ_0)$

$\delta(q_0, a, a) = (q_0, aaZ_0)$

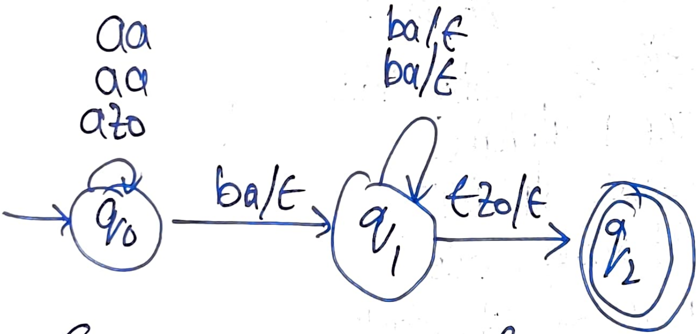
$\delta(q_0, a, a) = (q_0, aaaZ_0) \Rightarrow$

$\delta(q_0, b, a) = (q_1, \epsilon) =$

$\delta(q_1, b, a) = (q_1, \epsilon) =$

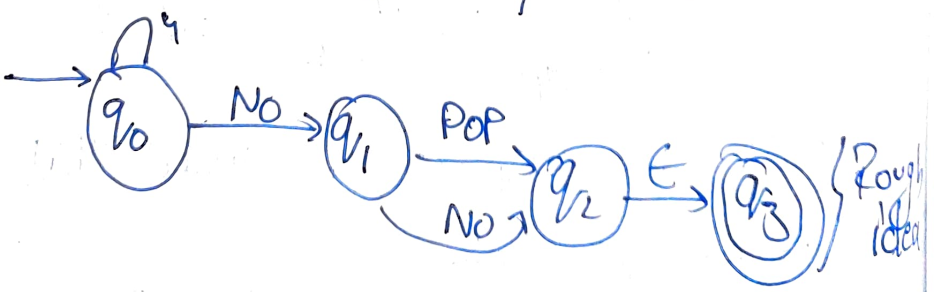
$\delta(q_1, b, a) = (q_1, \epsilon) =$

$\delta(q_1, \epsilon, Z_0) = (q_2, \epsilon) = \sqcup \rightarrow$ Empty Stack



Q- Construct the PDA for the following language
$L = \{a^n b^{2n} \mid n \geq 1\}$

$L = \{abb, aabbbb - - -\}$



a	a	b	b	b	ε

$\delta(q_0, a, Z_0) = (q_0, aZ_0)$

$\delta(q_0, a, a) = (q_0, aaZ_0)$

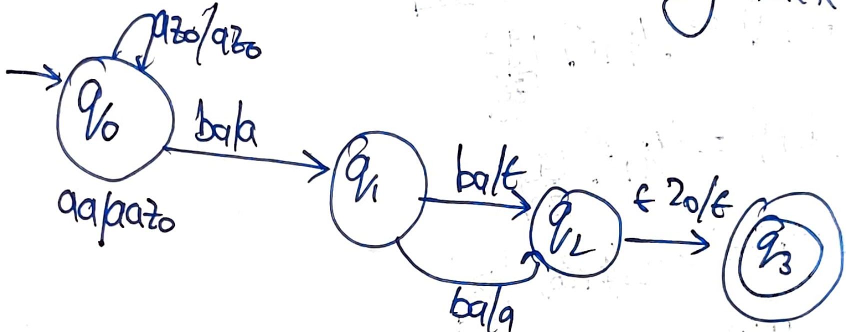
$\delta(q_0, b, a) = (q_1, a)$

$\delta(q_1, b, a) = (q_2, \epsilon)$

$\delta(q_2, b, a) = (q_1, a)$

$\delta(q_1, b, a) = (q_2, \epsilon)$

$\delta(q_2, \epsilon, Z_0) = (q_3, \epsilon)$ Empty stack



- Instantaneous Description of PDA

It's a formal notation to Explain how PDA Computes String

General Syntax

ID (q, w, δ)

$q \rightarrow$ Current State

$w \rightarrow$ I/6

$\delta \rightarrow$ Stack Content

Ex - ID $(q_0, aaabbb, Z_0)$

$(q_0, aabb, aZ_0)$

$(q_0, abbb, aaz_0)$

(q_0, bbb, aaz_0)

(q_1, bb, aaz_0)

(q_1, ϵ, z_0)

$(q_2, \epsilon\epsilon)$

★ Geometrical Representation of PDA

$M = (Q \Sigma \Gamma O\ q_0\ z_0\ F)$

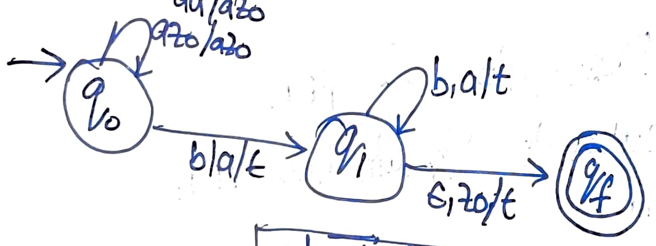
$\delta(q_0\ a\ z_0) = (q_0\ az_0)$ $\boxed{z_0}$

$\delta(q_0, a, a) = (q_0, aaz_0)$ $\boxed{\begin{matrix} a \\ z_0 \end{matrix}}$

$\delta(q_0, b, a) = (q_1, \epsilon)$ $\boxed{\begin{matrix} a \\ a \\ z_0 \end{matrix}}$

$\delta(q_1, b, a) = (q_1, \epsilon)$ $\boxed{\begin{matrix} a \\ z_0 \end{matrix}}$

$\delta(q_1, \epsilon, z_0) = (q_f, z_0)$ $\boxed{z_0}$



a	a	b	b	ε

Q $L = \{wcw^c \mid w \in (a+b)^+\}$

$L = \{abacaba, ababc - - - \}$

$\delta(q_0, a, z_0) = (q_0, az_0)$

$\delta(q_0, b, z_0) = (q_0, bz_0)$

$\delta(q_0, a, a) = (q_1, z_0 a a)$

$\delta(q_0, a, b) = (q_0, abz_0)$

$\delta(q_0, b, a) = (q_0, baz_0)$

$\delta(q_0, b, b) = (q_0, bbz_0)$

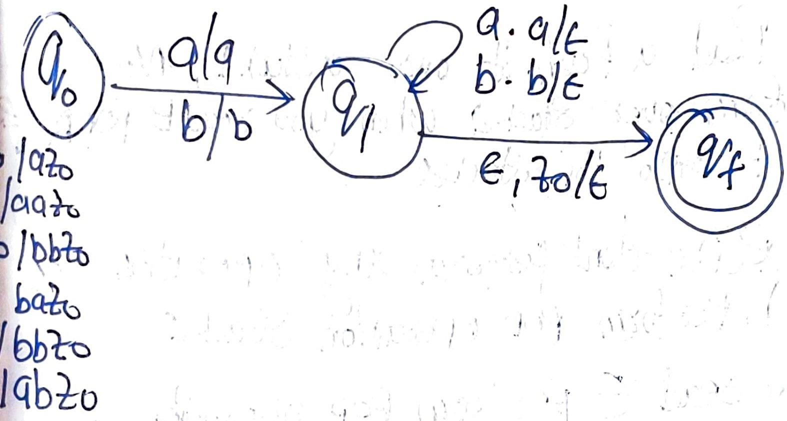
$\delta(q_0, c, a) = (q_1, a)$

$\delta(q_0, c, b) = (q_1, b)$

$\delta(q_1, a, a) = (q_2, \epsilon)$

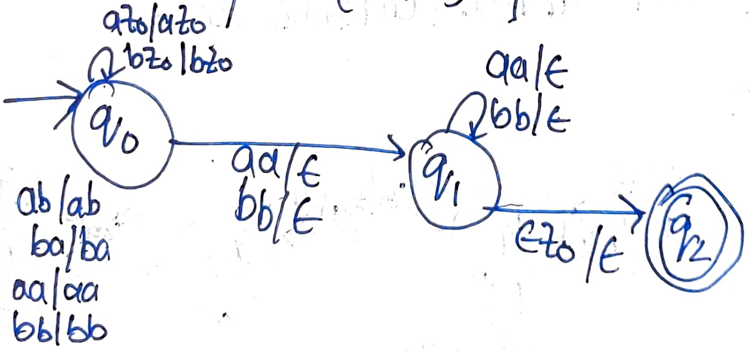
$\delta(q_1, b, b) = (q_2, \epsilon)$

$\delta(q_1, \epsilon, z_0) = (q_f, \epsilon)$



→ Ex of NDPDA

$L = \{ww^R \mid w \in (a+b)^+\}$



$\delta(q_0, az_0) = (q_0, az_0)$

$\delta(q_0, bz_0) = (q_0, bz_0)$

HW $\delta(q_0, ab) = (q_0, ab)$

$\delta(q_0, ba) = (q_0, ba)$

$\delta(q_0, aa) = (q_0, aa)$

$\delta(q_0, bb) = (q_0, bb)$

$\delta(q_1, aa) = (q_1, \epsilon)$

$\delta(q_1, bb) = (q_1, \epsilon)$

$\delta(q_0, aa) = (q_1, \epsilon)$

$\delta(q_0, bb) = (q_1, \epsilon)$

$\delta(q_1, \epsilon) = (q_2, \epsilon)$

★ 2 Stack PDA

(*) when you Read a Push it onto a Stack 1, No operation Perform over Stack 2 when you read B Pop A from Stack 1 and B onto the stack 2

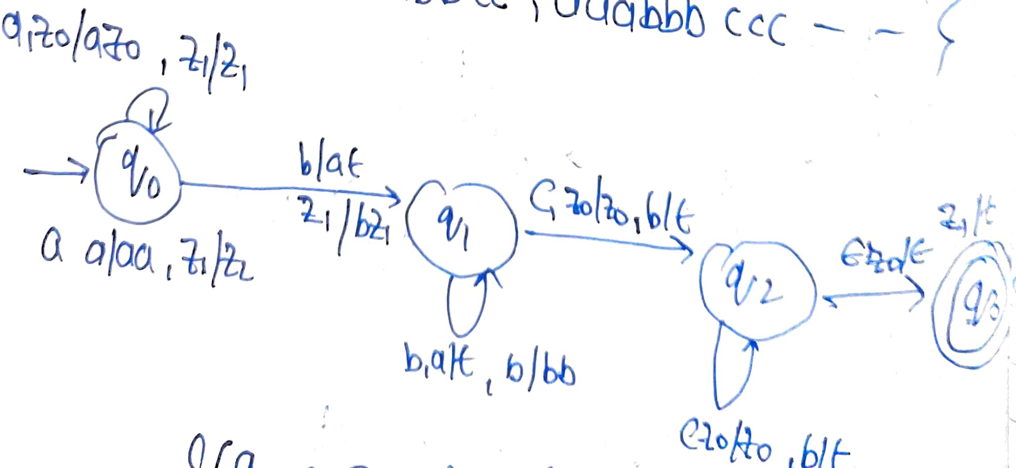
(*) when you Read c dont perform any operations on Stack 1 and Perform Pop operation Stack 2

(*) When you read ϵ Perform Pop operation on Both Stack

Imp
Ex

$L = \{a^n b^n c^n \mid n \geq 1\}$

$L = \{abc, aabbcc, aaabbbccc - - \}$



$\delta(q_0, a, z_0, z_1) = (q_0, az_0, z_1)$

→ **Conversion of CFG to PDA**

Let $G = (V T P S)$ be Context free Grammer, Construct PDA Accepting by Empty Stack

$\boxed{P = (Q \Sigma \Gamma \delta q_0 Z_0)}$ Traditional way

$P = (\{q\}\ T,\ V \cup T,\ \delta,\ q,\ S)$

(1) Transition δ Must use 2 Rules

(*) for Variable a(δ) of

$\delta(q, \epsilon, A) = \{q, B\}$ if $A \rightarrow B$ in R (Push opt)

(*) for Each Terminal a

$\delta(q, a, a) = (q, \epsilon)$ (Pop opt)

OM Ex- Construct a PDA for the following Grammer and test wheater 'abbabb' is N(P) language accepted by PDA by Each Empty Stack.

$G = (\{SA\}, \{ab\} R, S)$

$R = \{S \rightarrow AA / a$

$A \rightarrow SA / b\}$

$P = (\{q\}, \{a, b\}, \{SA, a, b\}, \delta, q, S)$

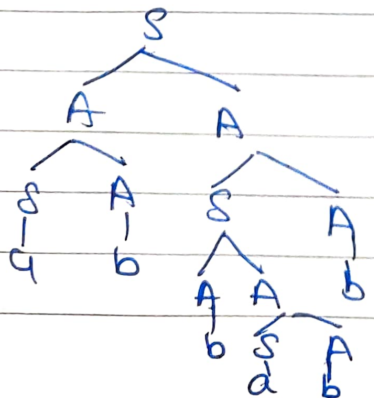
$\delta(q \epsilon S) = \{(q, AA) (q, a)\}$ — (1)

$\delta(q \epsilon A) = \{(q, SA), (q, b)\}$ — (2)

$\delta(q, aa) = (q, \epsilon)$ — (3)

$\delta(q, bb) = (q, \epsilon)$ — (4)

ow,



abbabb is accepted by CFG

DATE :

By Using (ID) Instantionous discription 'abbabb' is accepted by PDA by Empty Stack

N(P)

(q, abbabb, S)

$\vdash$ (q, abbabb, AA)

$\vdash$ (q, abbabb, SAA)

$\vdash$ (q, abbabb, aAA)

$\vdash$ (q, bbabb, AA)

$\vdash$ (q, bb,abb, bA)

$\vdash$ (q, babb, A)

$\vdash$ (q, babb, SA)

$\vdash$ (q, babb, AAA)

$\vdash$ (q, babb, bAA)

$\vdash$ (q, abb, AA)

$\vdash$ (q, abb, SAA)

$\vdash$ (q, abb, aAA)

$\vdash$ (q, bb, AA)

$\vdash$ (q, bb, bA)

$\vdash$ (q, b, A)

$\vdash$ (q, b, b)

$\vdash$ (q, ε, ε)